Mini Assessment

Name	Gorantla Venkata Hima Sandeep
Registration No.	99230041051
Year	IV
Department	CSE(AIML)

1. If an API returns 429, what should your program do?

A 429 status code means that the API has received too many requests.

The program should avoid sending more requests immediately.

It should wait for a short period before trying the request again.

A retry delay can be increased gradually using exponential backoff.

If the API provides a Retry-After value, the program should follow it.

The program should also limit repeated retries and handle the error safely.

2. If response.json() fails, what should your program do?

The program should not assume that every API response contains valid JSON.

It should use exception handling when converting the response to JSON.

If JSON parsing fails, the error should be caught instead of crashing the program.

The program can inspect the response as text to understand what was returned.

It should log the problem and provide an appropriate fallback or error message.

This makes the application more reliable when an API returns unexpected data.

3. If a JSON key is missing, why might .get() be safer than []?

The [] operator expects the requested JSON key to be present.

If the key is missing, using [] can raise a KeyError.

The .get() method can safely access a key without immediately raising an error.

When the key is absent, .get() normally returns None.

A default value can also be supplied, such as data.get('name', 'Unknown').

Therefore, .get() is safer when a JSON response may have missing keys.